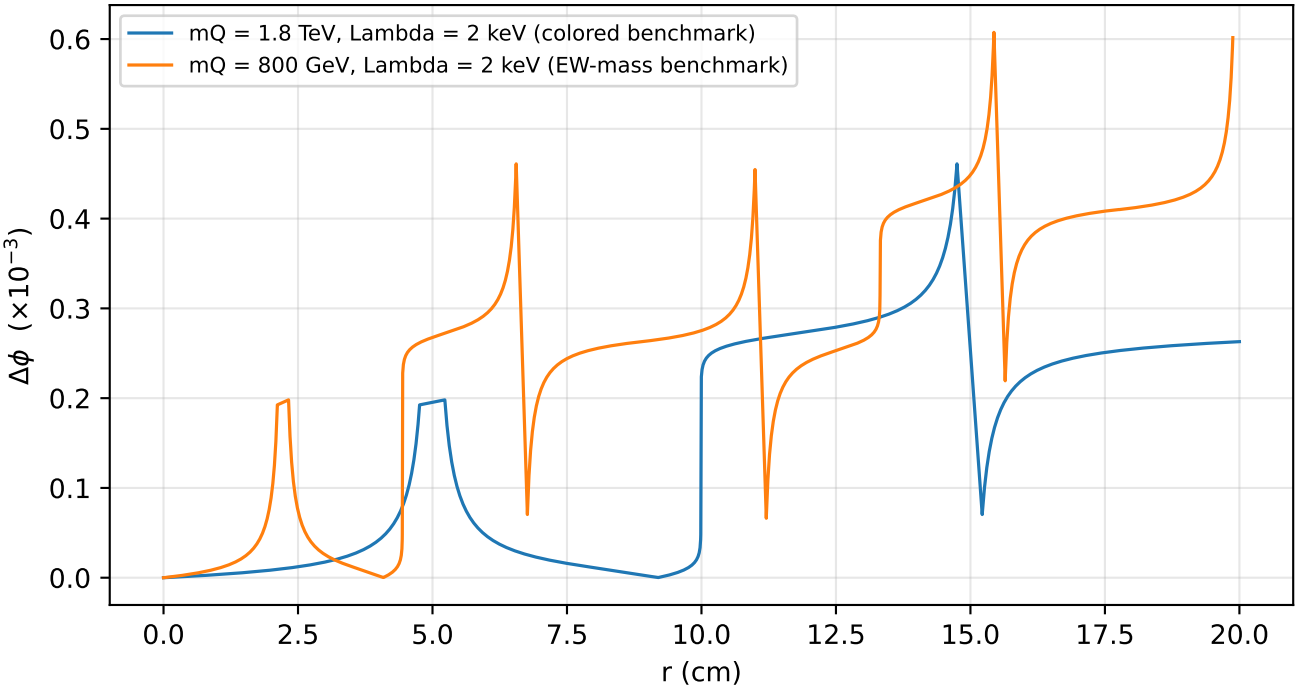
Figure 3 diagnostic reproduction (v2) -- FINAL APPROXIMATE / diagnostic plane-normal rotation of quirk velocity vectors vs CM radial distance (reproduced cache-only, from exported CSV values, not re-simulated)



Needle-like spikes arise from residual angle-unwrapping near plane-normal degeneracies; the overall trend and scale are interpreted qualitatively.